EC 571: Digital VLSI Circuit Design

CMOS Inverter Design and Analysis

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Acknowledgement - Some slides adapted from:

- J. Rabaey et al. Digital Integrated Circuits 2nd Ed.
- UC Berkeley EE141, UC Berkeley EE241, UC Berkeley EE251
- Stanford EE271

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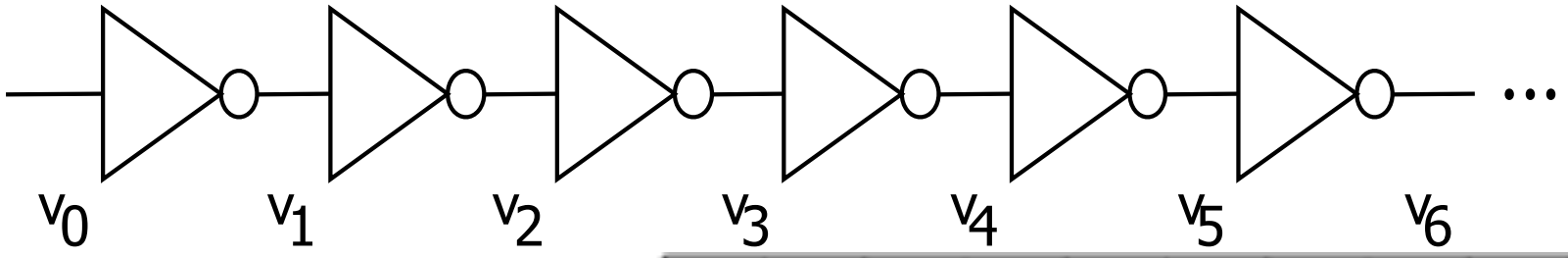
Last lecture

- Last lecture
  - Introduction to the field of Digital VLSI Circuits
  - Manufacturing
- Today's lecture
  - Switches
  - Inverters
  - Noise margins
- Reading
  - Chapter 5

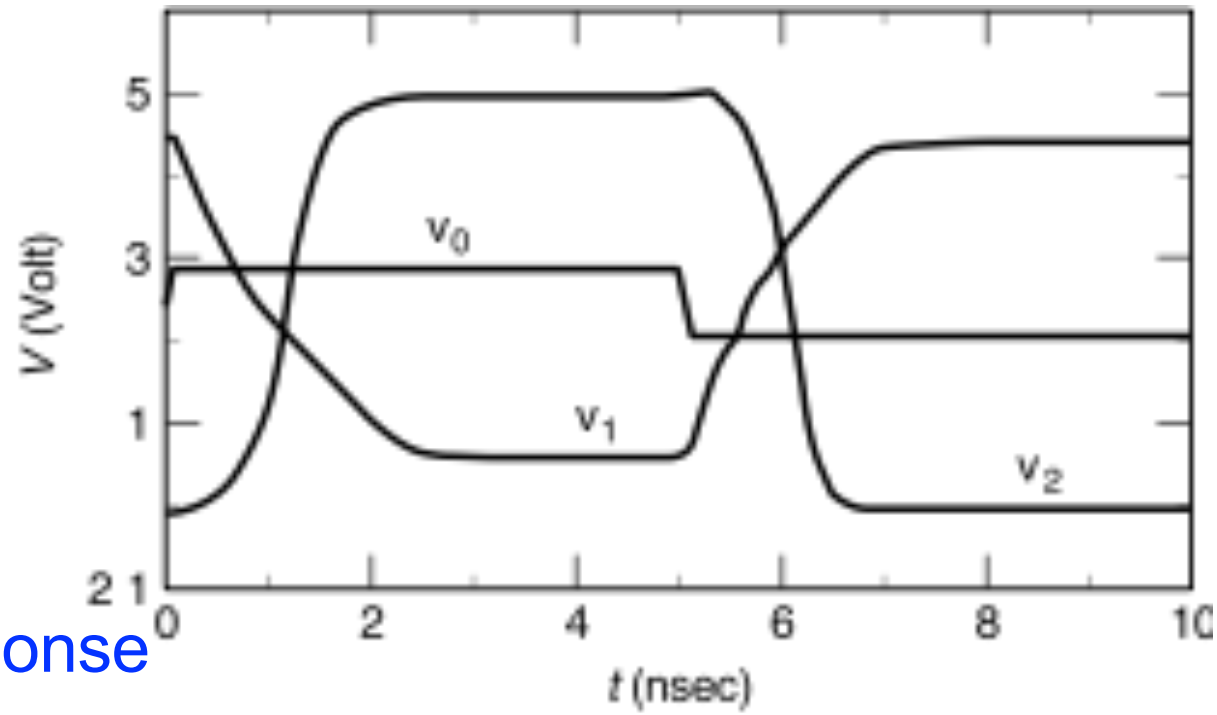
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Regenerative Property

Even if the input voltage/signal is forced into the noise region, the distributed signal converges back to the nominal voltage.



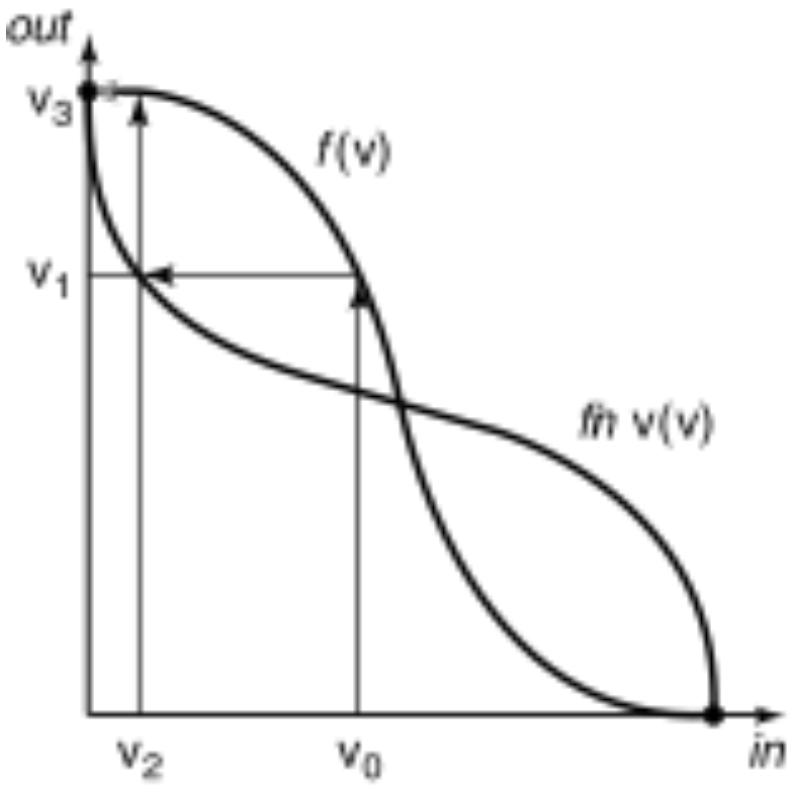
A chain of inverters



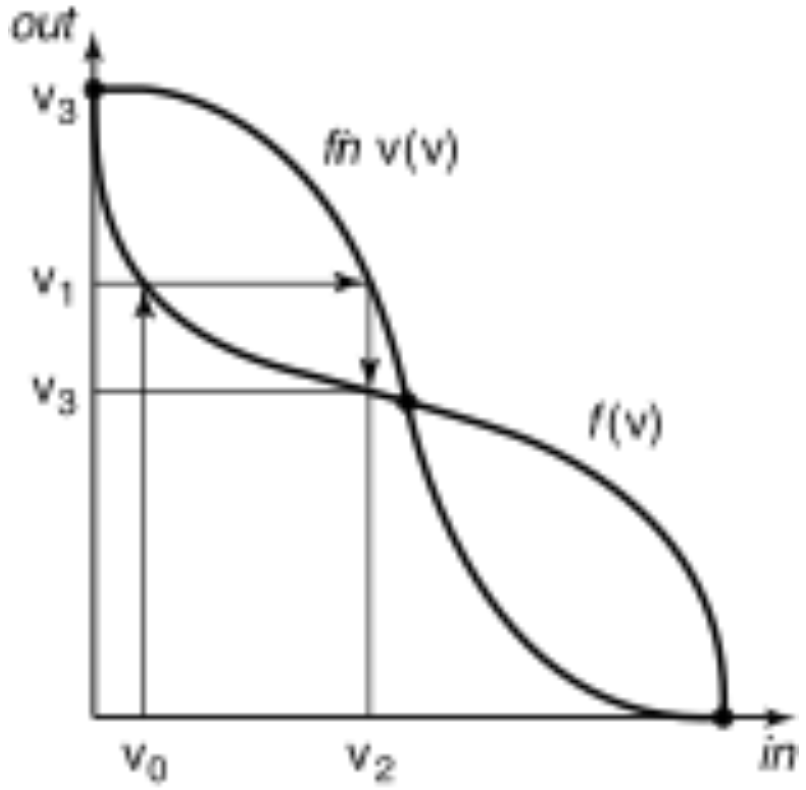
Simulated response

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Regenerative Property



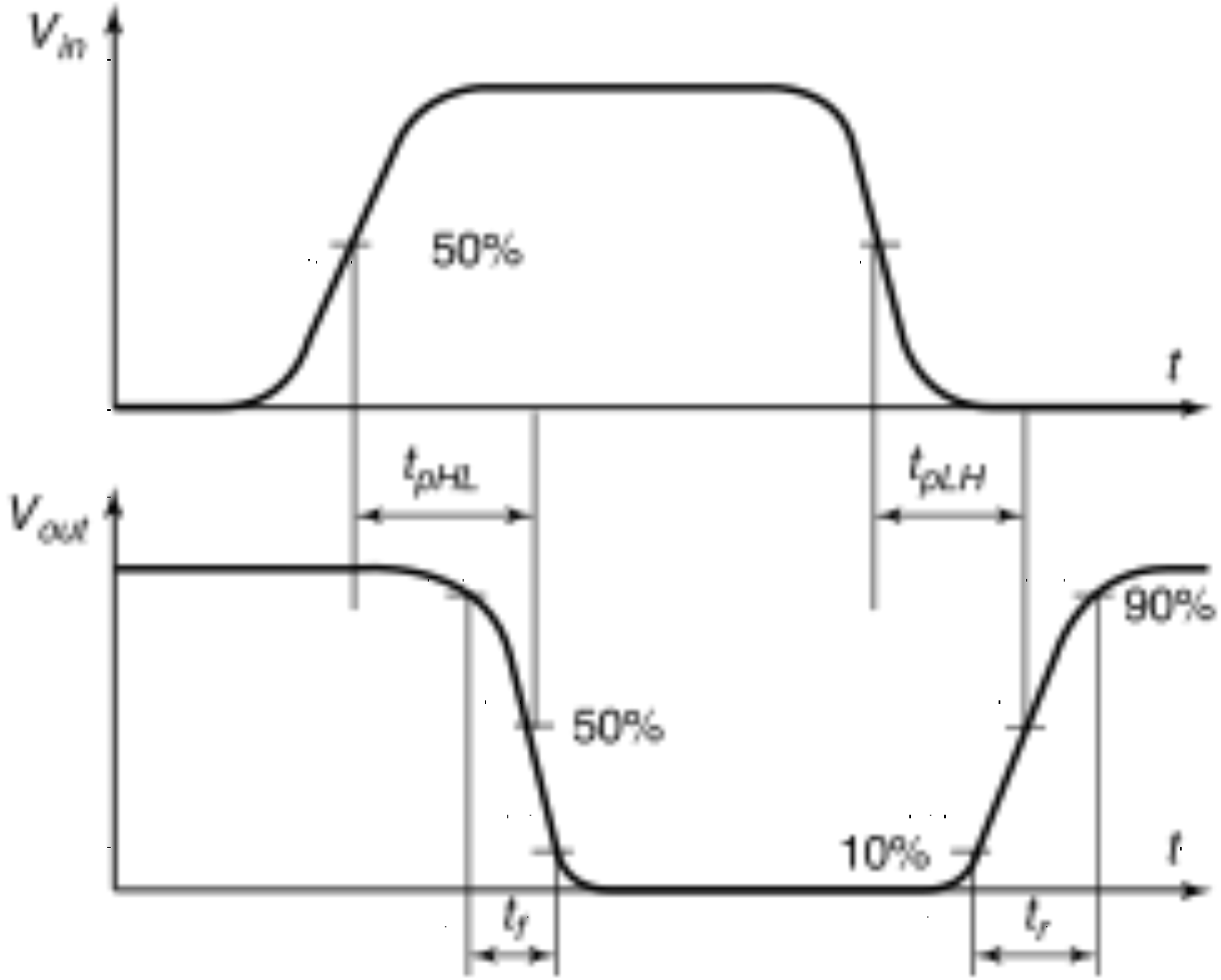
Regenerative



Non-Regenerative

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Delay Definitions



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